

Entrega TS

Métodos Matemáticos Para La Física

Métodos matemáticos para física. Curso 24/25.

Problemas ODE Tema 6. 10 puntos.

Problema 1 (3 puntos)

Encontrar la transformada de Laplace de las siguientes funciones

- a) $3 + t^5 + \sin(\pi t)$
- b) $a + bt + ct^2$
- c) $A\cos(\omega t) + B\sin(\omega t)$
- d) $\cos^2(\omega t)$.
- e) $4(t+1)^2$
- f) $\sin(\omega(t-a))$

Problema 2 (3 puntos)

Encontrar la función inversión de las siguientes transformadas de Laplace

- a) $F(s) = \frac{s^2+s+1}{s^3+s}$.
- b) $F(s) = \frac{1}{s+1}$.
- c) $F(s) = \frac{1}{s^2+4s+8}$.
- d) $F(s) = \frac{4}{s^2-9}$.
- e) $F(s) = \frac{2s}{s^2-1}$.
- f) $F(s) = \frac{s+1}{s^2+2s+10} + \frac{3}{s^2+2s+10}$.

Problema 3 (4 puntos)

Encontrar la solución a la siguiente ecuación diferencial usando la transformada de Laplace

- a) $x''(t) + x(t) = \cos(2t), x(0) = 0, x'(0) = 1$.
- b) $x''(t) + x(t) = f(t), x(0) = 0, x'(0) = 0$, con $f(t) = 1$ si $1 \leq t < 5$ y 0 en cualquier otro caso.
- c) $x'' + x = u(t-1), x(0) = 0, x'(0) = 0$.
- d) $x''' + x = (t-1)^3 u(t-1), x(0) = 1, x'(0) = 0, x''(0) = 0$.

Problema 1 (3 puntos)

Encontrar la trasformada de Laplace de las siguientes funciones

- a) $3 + t^5 + \sin(\pi t)$
- b) $a + bt + ct^2$
- c) $A \cos(\omega t) + B \sin(\omega t)$
- d) $\cos^2(\omega t)$.
- e) $4(t+1)^2$
- f) $\sin(\omega(t-a))$

$$a) 3 + t^5 + \sin(\pi t) \quad \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad \mathcal{L}\{\sin(kt)\} = \frac{k}{s^2 + k^2}$$

$$\mathcal{L}\{3\} = 3 \mathcal{L}\{t^0\} = 3 \cdot \frac{0!}{s} = \frac{3}{s}$$

$$\mathcal{L}\{t^5\} = \frac{5!}{s^6} = \frac{120}{s^6}$$

$$\mathcal{L}\{\sin \pi t\} = \frac{\pi}{s^2 + \pi^2} \rightarrow \mathcal{L}\{3 + t^5 + \sin \pi t\} = \frac{3}{s} + \frac{120}{s^6} + \frac{\pi}{s^2 + \pi^2}$$

$$b) a + bt + ct^2 \rightarrow \mathcal{L}\{a + bt + ct^2\} = \frac{a \cdot 0!}{s} + \frac{b \cdot 1!}{s^2} + \frac{c \cdot 2!}{s^3} = \frac{a}{s} + \frac{b}{s^2} + \frac{2c}{s^3}$$

$$c) \mathcal{L}\{A \cos(\omega t) + B \sin(\omega t)\} = \frac{As}{s^2 + \omega^2} + \frac{B\omega}{s^2 + \omega^2} = \frac{As + B\omega}{s^2 + \omega^2}$$

$$d) \cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = \cos^2(\theta) - (1 - \cos^2(\theta)) = 2\cos^2(\theta) - 1$$

$$\cos^2(\theta) = \frac{\cos(2\theta) + 1}{2}$$

$$\mathcal{L}\{\cos^2(\omega t)\} = \frac{1}{2} \mathcal{L}\{\cos(2\omega t) + 1\} = \frac{1}{2} \left(\frac{s}{s^2 + 4\omega^2} + \frac{1}{s} \right) = \frac{s}{2s^2 + 8\omega^2} + \frac{1}{2s}$$

$$e) 4 \mathcal{L}\{(t+1)^2\} = 4 \mathcal{L}\{t^2 + 2t + 1\} = 4 \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right) = \frac{8}{s^3} + \frac{8}{s^2} + \frac{4}{s}$$

5) $\mathcal{L}\{\sin(\omega(t-a))\}$ Propiedad de Desplazamiento: $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$

$$\mathcal{L}\{\sin(\omega t - \omega a)\} = e^{-as} \cdot \frac{\omega}{s^2 + \omega^2}$$

Problema 2 (3 puntos)

Encontrar la función inversión de las siguientes transformadas de Laplace

a) $F(s) = \frac{s^2 + s + 1}{s^3 + s}$

b) $F(s) = \frac{1}{s+1}$

c) $F(s) = \frac{1}{s^2 + 4s + 8}$

d) $F(s) = \frac{4}{s^2 - 9}$

e) $F(s) = \frac{2s}{s^2 - 1}$

f) $F(s) = \frac{s+1}{s^2 + 2s + 10} + \frac{3}{s^2 + 2s + 10}$

a) $F(s) = \frac{s^2 + s + 1}{s^3 + s} = \frac{s^2 + s + 1}{s(s^2 + 1)}$

Descompongo en fracciones simples:

$$\frac{s^2 + s + 1}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + D}{s^2 + 1} \Rightarrow s^2 + s + 1 = A(s^2 + 1) + (Bs + D)s$$

caso $s = 0$: $A = 1$

$$s^2 + s + 1 = s^2 + 1 + Bs^2 + Ds = (B+1)s^2 + Ds + 1$$

$$\begin{aligned} (B+1)s^2 &= 1s^2 \rightarrow B = 0 \rightarrow \frac{s^2 + s + 1}{s^3 + s} = \frac{1}{s} + \frac{1}{s^2 + 1} \\ Ds &= s \rightarrow D = 1 \end{aligned}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s} + \frac{1}{s^2 + 1}\right\} = 1 + \sin(t)$$

$$b) \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} = e^{-t}$$

$$c) \frac{1}{s^2 + 4s + 8} = \frac{1}{(s+2)^2 + 2^2} = e^{-2t} \frac{\sin(2t)}{2}$$

$$d) \mathcal{L}^{-1} \left\{ \frac{4}{s^2 - 9} \right\} = \frac{4}{3} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 - 9} \right\} = \frac{4}{3} \sinh(3t) = \frac{\frac{4}{3}e^t - \frac{4}{3}e^{-t}}{2} = \frac{2}{3}e^t - \frac{2}{3}e^{-t}$$

$$e) \frac{2s}{s^2 - 1} = 2 \mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 1} \right\} = 2 \cosh(t) = 2 \cdot \frac{e^t + e^{-t}}{2} = e^t + e^{-t}$$

$$f) \frac{s+1}{s^2 + 2s + 10} + \frac{3}{s^2 + 2s + 10} = \frac{s+1}{(s+1)^2 + 9} + \frac{3}{(s+1)^2 + 9}$$

$$\mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2 + 9} \right\} = e^{-t} \cos(3t)$$

$$\mathcal{L}^{-1} \left\{ \frac{3}{(s+1)^2 + 9} \right\} = e^{-t} \sin(3t)$$

$$\mathcal{L}^{-1} \left\{ \frac{s+4}{s^2 + 2s + 10} \right\} = e^{-t} (\sin(3t) + \cos(3t))$$

Problema 3 (4 puntos)

Encontrar la solución a la siguiente ecuación diferencial usando la transformada de Laplace

- $x''(t) + x(t) = \cos(2t), x(0) = 0, x'(0) = 1.$
- $x''(t) + x(t) = f(t), x(0) = 0, x'(0) = 0$, con $f(t) = 1$ si $1 \leq t < 5$ y 0 en cualquier otro caso.
- $x'' + x = u(t-1), x(0) = 0, x'(0) = 0.$
- $x''' + x = (t-1)^3 u(t-1), x(0) = 1, x'(0) = 0, x''(0) = 0.$

$$a) \quad x''(t) + x(t) = \cos(2t) \quad x(0) = 0, x'(0) = 1$$

$$\mathcal{L}\{x''\} = s^2 X(s) - s x(0) - x'(0) = s^2 X(s) - 1$$

$$\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2 + 4}$$

$$s^2 X(s) - 1 + X(s) = \frac{s}{s^2 + 4} \rightarrow X(s)(s^2 + 1) = \frac{s}{s^2 + 4} + 1 = \frac{s^2 + s + 4}{s^2 + 4} \Rightarrow X(s) = \frac{s^2 + s + 4}{(s^2 + 4)(s^2 + 1)}$$

$$\frac{s^2 + s + 4}{(s^2 + 4)(s^2 + 1)} = \frac{As + B}{s^2 + 1} + \frac{Cs + D}{s^2 + 4} \rightarrow s^2 + s + 4 = (As + B)(s^2 + 4) + (Cs + D)(s^2 + 1) = As^3 + 4As + Bs^2 + 4B + Cs^3 + Cs + Ds^2 + D = (A+C)s^3 + (B+D)s^2 + (4A+C)s + (4B+D)$$

$$s^2 + s + 4 = (A+C)s^3 + (B+D)s^2 + (4A+C)s + (4B+D)$$

$$\left. \begin{array}{l} A+C = 0 \rightarrow A = -C \rightarrow A = 1/3 \\ 4A+C = 1 \rightarrow -4C+C=1 \rightarrow C = -1/3 \\ B+D = 1 \rightarrow D = 1-B \rightarrow D=0 \\ 4B+D = 4 \rightarrow 4B+1-B=4 \rightarrow B=1 \end{array} \right\} \quad \frac{s^2 + s + 4}{(s^2 + 4)(s^2 + 1)} = \frac{1/3 s + 1}{s^2 + 1} - \frac{1/3 s}{s^2 + 4} = \frac{1/3}{s^2 + 1} + \frac{1}{s^2 + 1} - \frac{1/3}{s^2 + 4}$$

$$\mathcal{L}^{-1}\{X(s)\} = 1/3 \cos(t) + \sin(t) - 1/3 \cos(2t) = f(t)$$

$$x(t) = 1/3 [\cos(t) + 3\sin(t) - \cos(2t)]$$

$$b) x''(t) + x(t) = f(t) \quad x(0) = 0, x'(0) = 0 \quad f(t) = \begin{cases} 1 & \text{si } 1 \leq t < 5 \\ 0 & \text{si resto} \end{cases}$$

$$\mathcal{L}\{x''\} = s^2 X(s) - s x(0) - x'(0) = s^2 X(s)$$

Función escalón unitario: $f(t) = u(t-1) - u(t-5)$

$$\mathcal{L}\{u(t-1) - u(t-5)\} = \frac{e^{-s}}{s} - \frac{e^{-5s}}{s}$$

$$s^2 X(s) + X(s) = \frac{e^{-s}}{s} - \frac{e^{-5s}}{s} \rightarrow X(s) = \frac{e^{-s}}{s(s^2+1)} - \frac{e^{-5s}}{s(s^2+1)}$$

Descomponemos $\frac{1}{s(s^2+1)}$

$$\frac{1}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1}$$

$$1 = A(s^2+1) + (Bs+C)s = As^2 + A + Bs^2 + Cs = (A+B)s^2 + (C+A)$$

$$A=1, B=-1, C=0 \rightarrow \frac{1}{s} - \frac{s}{s^2+1} = \frac{1}{s(s^2+1)} \quad \mathcal{L}^{-1}\{X(s)\} = 1 - \cos(t)$$

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = u(t-a) f(t-a)$$

$$\mathcal{L}^{-1}\left\{e^{-s} \cdot \frac{1}{s(s^2+1)}\right\} = u(t-1)(1 - \cos(t-1)) \quad , \quad \mathcal{L}^{-1}\left\{e^{-5s} \cdot \frac{1}{s(s^2+1)}\right\} = u(t-5)(1 - \cos(t-5))$$

$$x(t) = u(t-1)(1 - \cos(t-1)) - u(t-5)(1 - \cos(t-5))$$

$$c) x'' + x = u(t-1), \quad x(0) = 0, \quad x'(0) = 0$$

$$\mathcal{L}\{x''\} = s^2 X(s), \quad \mathcal{L}\{u(t-1)\} = \frac{e^{-s}}{s}$$

$$s^2 X(s) + X(s) = \frac{e^{-s}}{s} \rightarrow X(s) = \frac{e^{-s}}{s(s^2+1)}$$

Descompongo $\frac{1}{s(s^2+1)} = \frac{1}{s} - \frac{s}{s^2+1} \rightarrow \mathcal{L}^{-1}\left\{e^{-s} \cdot \frac{1}{s(s^2+1)}\right\} = u(t-1)(1 - \cos(t-1))$

$$x(t) = u(t-1)(1 - \cos(t-1))$$

$$d) x''' + x = (t-1)^3 u(t-1), \quad x(0)=1, \quad x'(0)=0, \quad x''(0)=0$$

$$\mathcal{L}\{x'''\} = s^3 X(s) - s^2, \quad \mathcal{L}\{(t-1)^3 u(t-1)\} = e^{-s} \frac{6}{s^4}$$

$$s^3 X(s) - s^2 + X(s) = \frac{6e^{-s}}{s^4}$$

$$(s^3 + 1)X(s) = \frac{6e^{-s}}{s^4} + s^2 \rightarrow X(s) = \frac{6e^{-s}}{s^4(s^3+1)} + \frac{s^2}{(s^3+1)}$$

$$(s^3+1) = (s+1)(s^2-s+1)$$

$$\text{Descomposición } \frac{6}{s^4(s^3+1)}$$

$$\frac{6}{s^4(s^3+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s^4} + \frac{E}{s+1} + \frac{Fs+G}{s^2-s+1}$$

$$6 = A s^3(s^3+1) + B s^2(s^3+1) + C s(s^3+1) + D(s^3+1) + E s^4(s^2-s+1) + (Fs+G) \frac{s^4(s+1)}{(s^5+s^4)}$$

$$6 = (A+E+F)s^6 + (B+F+G-E)s^5 + (C+E+G)s^4 + (D+A)s^3 + Bs^2 + (s+D)$$

$$\text{caso } s=0: D=6 \quad B=0=C$$

$$\text{caso } s=-1: 3E=6 \rightarrow E=2$$

$$A+F+2=0 \rightarrow -6+F=-2 \rightarrow F=4$$

$$A=-6, B=0, C=0, D=6, E=2, F=4, G=2$$

$$F+G-2=0 \rightarrow \text{Se cumple}$$

$$2+G=0 \rightarrow G=-2$$

$$A+6=0 \rightarrow A=-6$$

$$\frac{6}{s^4(s^3+1)} = -\frac{6}{s} + \frac{6}{s^4} + \frac{2}{s+1} + \frac{4s-2}{(s^2-s+1)}$$

$$\text{Descompongo: } \frac{s^2}{(s+1)(s^2-s+1)} = \frac{A}{s+1} + \frac{Bs+C}{s^2-s+1} = A(s^2-s+1) + (Bs+C)(s+1)$$

$$\text{caso } s=-1: 1 = 3A \rightarrow A = \frac{1}{3} \rightarrow \frac{1}{3}s^2 - \frac{1}{3}s + \frac{1}{3} + Bs^2 + (B+C)s + C$$

$$B + \frac{1}{3} = 1 \rightarrow B = \frac{2}{3}$$

$$B+C - \frac{1}{3} = 0 \rightarrow C + \frac{2}{3} = \frac{1}{3} \rightarrow C = -\frac{1}{3}$$

$$\frac{s^2}{s^3+1} = \frac{1}{s} \cdot \frac{1}{s+1} + \frac{1}{s} \cdot \frac{2s-1}{s^2-s+1}$$

$$X(s) = e^{-s} \left(-\frac{6}{s} + \frac{6}{s^4} + \frac{2}{s+1} + \frac{4s-2}{(s-\frac{1}{2})^2 + \frac{3}{4}} \right) + \frac{1}{s} \cdot \frac{1}{s+1} + \frac{2}{s} \cdot \frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2 + \frac{3}{4}}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s} \cdot \frac{1}{s+1} + \frac{2}{s} \cdot \frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2 + \frac{3}{4}} \right\} = \frac{1}{3} e^{-t} + \frac{2}{3} e^{\frac{t}{2}} \cos\left(\frac{\sqrt{3}}{2}t\right)$$

$$\mathcal{L}^{-1} \left\{ -\frac{6}{s} + \frac{6}{s^4} + \frac{2}{s+1} + 4 \frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2 + \frac{3}{4}} \right\} = -6 + t^3 + 2e^{-t} + 4e^{\frac{t}{2}} \cos\left(\frac{\sqrt{3}}{2}t\right)$$

$$\mathcal{L}^{-1} \left\{ e^{-s} \left(-\frac{6}{s} + \frac{6}{s^4} + \frac{2}{s+1} + 4 \frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2 + \frac{3}{4}} \right) \right\} = u(t-1) \left(-6 + (t-1)^3 + 2e^{-(t-1)} + 4e^{\frac{t-1}{2}} \cos\left(\frac{\sqrt{3}}{2}(t-1)\right) \right)$$

$$f(t) = u(t-1) \left(-6 + (t-1)^3 + 2e^{-(t-1)} + 4e^{\frac{t-1}{2}} \cos\left(\frac{\sqrt{3}}{2}(t-1)\right) \right) + \frac{1}{3}e^{-t} + \frac{2}{3}e^{\frac{t}{2}} \cos\left(\frac{\sqrt{3}}{2}t\right)$$